

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Computer Vision with OpenCV COURSE CODE: DSA02

COURSE OUTCOME COVERED

CO5: Apply computer vision techniques to real-world applications such as face recognition, tracking, medical imaging, and autonomous systems. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5 Total Marks:50 Annexure A

Constraint-Based Problem Solving

Code of Conduct:

*I _JAASIEL J S (192524046)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly	
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization	
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints	
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts	

Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided	
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Annexure A

Constraint-Based Problem Solving

1. Scenario: A company wants to deploy a secure access system using face recognition in a low-light environment.

Constraints:

- A. Limited computational resources (embedded device)
- B. Faces may appear at different orientations ($\pm 30^\circ$)
- C. Real-time recognition required (<200 ms per face)

Questions:

1. How would you adapt Eigenface-based recognition to handle varying illumination and orientation?

Ans: Use illumination normalization and train Eigenfaces with multiple face orientations. Apply face alignment to handle rotations up to $\pm 30^\circ$.

2. Suggest preprocessing steps to improve recognition under low-light conditions.

Ans: Use histogram equalization or CLAHE, noise reduction, gamma correction, and image normalization.

3. Discuss strategies to optimize computation for real-time performance on constrained hardware.

Ans: Reduce image resolution, use fewer Eigenfaces, optimize matrix operations, and use hardware acceleration where available.

Question 2: Photo Album Organization

2. Scenario: You are designing an AI system to automatically organize a large personal photo collection by people and events.

Constraints:

- A. Millions of images with inconsistent resolution and quality
- B. Multiple photos per person in different poses
- C. Privacy considerations: all processing must occur locally

Questions:

1. How would you use face recognition and clustering to group photos of the same individual?

Ans: Detect faces, extract facial features, compare face embeddings, and use clustering to group photos of the same person.

2. Propose a method to detect and separate images with occlusions or partial faces.

Ans: Use face detection with facial landmark and visibility checks. Images with a small visible face area can be classified as occluded or partial faces.

3. Discuss trade-offs between accuracy and computational efficiency for local processing.

Ans: Higher accuracy generally requires more computation, while lightweight models reduce computation and protect privacy but may slightly reduce accuracy.

3. Scenario: A smart city surveillance system must detect moving objects and track pedestrians across multiple cameras.

Constraints:

- A. Crowded scenes with frequent occlusion**
- B. Limited network bandwidth between cameras**
- C. Real-time processing requirement**

Questions:

1. How would you design a foreground–background separation algorithm to minimize false positives in crowded scenes?

Ans: Use adaptive background modeling, morphological filtering, and shadow suppression to separate foreground objects and reduce false positives.

2. Explain how particle filters can be used to handle occlusion during tracking.

Ans: Particle filters maintain multiple possible object positions and update them using motion and observation probabilities, allowing tracking to continue during temporary occlusion.

3. Propose an efficient multi-camera strategy for consistent pedestrian identification across viewpoints.

Ans: Use lightweight person detection, feature-based re-identification, and camera-to-camera feature matching to maintain consistent pedestrian identities.

4. Scenario: A self-driving car must navigate urban streets while detecting lanes, road signs, and pedestrians.

Constraints:

- A. Variable weather and lighting conditions (rain, night)
- B. Real-time decision-making (≤ 50 ms per frame)
- C. High-density traffic with moving pedestrians

Questions:

1. How would you implement road detection robust to shadows, rain, and nighttime illumination?

Ans: Use color and edge-based lane detection with adaptive thresholding, shadow removal, perspective transformation, and illumination normalization.

2. Propose a pipeline to simultaneously detect road signs and pedestrians efficiently.

Ans: Capture and preprocess the frame, detect the road region, and use lightweight object detectors in parallel for road signs and pedestrians.

3. Discuss the trade-offs between deep learning-based detectors and classical computer vision methods under real-time constraints.

Ans: Deep learning provides better accuracy and robustness but requires more computation, while classical computer vision is faster and lighter but less robust in complex conditions.

5. Scenario: An AR system for surgical assistance must overlay segmented organs in real time.

Constraints:

- A. High-resolution 3D medical images (CT/MRI)
- B. Latency must be < 100 ms for surgeon interaction
- C. Limited GPU memory on wearable AR headset

Questions:

1. How would you implement real-time organ segmentation under these memory and latency constraints?

Ans: Use a lightweight segmentation network with cropped regions of interest, reduced-resolution processing, and GPU acceleration to meet the latency requirements.

2. Discuss methods to integrate segmentation results accurately in AR/MR overlays.

Ans: Register the 3D segmentation with the patient's coordinate system using image registration, depth information, and spatial tracking for

accurate AR/MR overlay.

3. Suggest strategies to reduce computational load while maintaining segmentation accuracy.

Ans: Use model compression, quantization, pruning, reduced image resolution, region-of-interest processing, and lightweight network architectures.